

Unit: Unit 8: Quadratics

Topic: Graphing Quadratics and Properties of Parabolas

Name: _____

Date: _____

Quiz (Carolina Reaper)

Open Ended Questions (FRQ)

Q9. The distance d that a truck travels is related to the time t it travels by the equation $d = -t^2 + 12t + 10$. Graph the equation, and find the vertex, axis of symmetry, and t -intercepts. Show all work and explain your reasoning.

Q10. A shipping company wants to minimize the cost of delivering packages. The cost function is given by $C(x) = x^2 - 8x + 20$, where x is the number of packages. Find the minimum cost and the number of packages that results in this minimum cost. Show all work and explain your reasoning.

Chef's Tips

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- Emphasize the importance of showing work and explaining reasoning.
- Encourage students to use graphing calculators to visualize the parabolas.
- Remind students to check their units and ensure they are consistent.